Hee Han

Researcher, Insect Systematics and Evolution Laboratory
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ResearchGate ORCiD Personal Website

Current Position

Researcher, Insect Systematics and Evolution Lab, Jeonbuk National University, Korea. 2026–present

Education

M.S.A., Jeonbuk National University, Korea 2024–2025

Systematic Study of the Subfamily Erebinae (Lepidoptera: Erebidae) from the Korean Peninsula with Phylogenomics.

Advisor: Prof. Dr. Sora Kim

B.S.A., Jeonbuk National University, Korea 2018–2024

Experiences

Researcher, Insect Systematics and Evolution Laboratory, Jeonbuk National University 2026–present

Advisor: Prof. Dr. Sora Kim

Visiting Researcher, The Kawahara Lab, University of Florida 2025

Principal Collaborator: Prof. Dr. Akito Y. Kawahara

Researcher, Lab. of Insect Phylogeny and Evolution, Jeonbuk National University 2024–2025

Advisor: Prof. Dr. Sora Kim

Undergraduate student, Lab. of Insect Phylogeny and Evolution, Jeonbuk National University 2023–2024

Advisor: Prof. Dr. Sora Kim

Visiting Researcher, Natural History Museum of Jeonbuk National University 2021–2022

Principal Collaborator: Ms. Ji-in Yang

Undergraduate student, Insect Taxonomy Lab, Jeonbuk National University 2021–2022

Advisor: Prof. Dr. Yong Hong

Honors, Awards & Fellowships

Graduate Research Assistant Fellowship, Jeonbuk National University 2025

Advisor: Prof. Dr. Sora Kim

Excellence Oral communication Prize, 2025 Fall International Conference of Korean Society of Applied Entomology (Jeju, South Korea) 2025

Brain Korea 21 (BK21) Graduate Research Fellowship: funded by the NRF 2024–present

Excellence Poster Presentation Prize, 2024 Autumn Conference of Korean Society of Applied Entomology (Hwaseong, South Korea) 2024

Excellence Insect Photoraphy Prize, 2024 Autumn Conference of Korean Society of Applied Entomology (Hwaseong, South Korea) 2024

University Presidenet's Honors for Academic Excellence, Jeonbuk National University 2022

Publications

6. **Han, H.**, Martinez, J.I., Plotkin, D., Masonick, P.K., Bethin, J., Ravelomanana, A., Kawahara, A.Y., Kim, S. A comprehensive dated phylogeny of Erebidae (Lepidoptera, Noctuoidea) clarifies evolutionary relationships, divergence times, and taxonomy. Systematic Entomology (under review).

5. **Han, H.**, Park, D.H., Kim, S. (2026) A new record of the southern erebid moth species, *Hypena nakajimai* Kishida, 2010 (Lepidoptera, Erebidae), from Korea with molecular analyses. Biodiversity Data Journal, 14:e198224. DOI: https://doi.org/10.3897/BDJ.14.e198224

4. Cha, Y.-B., **Han, H.**, Ra, D.-K., Kim, S. (2026) A new species of *Dudusa* (Lepidoptera: Notodontidae) from the Oriental Region with an annotated checklist of the genus. Zootaxa, 5821 (4), 571-580. DOI: https://doi.org/10.11646/zootaxa.5821.4.9

3. **Han, H.** & Kim, S. (2025) A new subspecies of *Avitta fasciosa* Moore, 1882 (Lepidoptera, Erebidae) from Korea. Biodiversity Data Journal, 13: e176317. DOI: https://doi.org/10.3897/BDJ.13.e176317

2. **Han, H.** & Kim, S. (2025) First records of genera *Chrysopera* Hampson, 1894 and *Entomogramma* Guenée, 1852 of the subfamily Erebinae (Lepidoptera, Erebidae) from South Korea. Biodiversity Data Journal,, 13:e139471. DOI: https://doi.org/10.3897/BDJ.12.e139471

1. **Han, H.**, Ulzijargal, B., Kim, S. (2024) Taxonomic Review of the Genus *Thyas* Hübner (Lepidoptera: Noctuoidea: Erebidae) from Korea, including the First Description of *T.coronata.* Korean Journal of Applied Entomology, 63(1): 5-11. DOI: https://doi.org/10.5656/KSAE.2024.01.1.048

Scientific Communications

10. **Han, H.**, Martinez, J.I., Plotkin, D., Masonick, P.K., Bethin, J., Ravelomanana, A., Kawahara, A.Y., Kim, S. (2026) Anchored hybrid enrichment reveals the evolutionary history of the largest moth family (Lepidoptera: Noctuoidea: Erebidae). 2026 Fall International Conference of Korean Society of Applied Entomology (Buan, South Korea). [Poster]

9. Jeong, K.-H., **Han, H.**, Park, J., Kim, J., Johnson, J., Lin, H.-Y., Harms, D., Kim., S. (2026) Comprehensive phylogenetics of the Pseudoscorpiones using backbone and data-mining. 2026 Spring Conference of Korean Society of Applied Entomology & The Entomological Society of Korea (Cheonju, South Korea). [Poster]

8. **Han, H.**, Plotkin, D., Masonick, P. K., Kawahara, A. Y., & Kim, S. (2025) Phylogenomics of the family Erebidae (Lepidoptera, Noctuoidea). 2025 Fall International Conference of Korean Society of Applied Entomology (Jeju, South Korea). [Oral communication]

7. **Han, H.** & Kim, S. (2025) Revealing evolutionary trajectories of the family Erebidae (Insecta: Lepidoptera) based on existing phylogeny. The 80th Annual Meeting of the Korean Association of Biological Sciences (Seoul, South Korea). [Poster]

6. **Han, H.** & Kim, S. (2025) First records of two species within the subfamily Erebinae (Lepidoptera, Erebidae) from the Korean Peninsula. Spring Conference of Korean Society of Applied Entomology & The Entomological Society of Korea 2025 (Hongcheon, South Korea). [Poster]

5. **Han, H.** & Kim, S. (2024) Pictorial key of underwing moths (Lepidoptera: Erebidae: *Catocala*) from Korean Peninsula. Autumn Conference of Korean Society of Applied Entomology 2024 (Hwaseong, South Korea). [Poster]

4. **Han, H.** & Kim, S. (2024) Species diversity comparisons of subtropical and tropical Lepidoptera in Vietnam: Noctuoidea fauna and flora. Spring Conference of Korean Society of Applied Entomology & The Entomological Society of Korea 2024 (Cheongju, South Korea). [Poster]

3. **Han, H.** & Kim, S. (2023) A historical review of the classification of subfamily Erebinae (Lepidoptera: Erebidae) in Korea. Autumn Conference of Korean Society of Applied Entomology 2023 (Pyeongchang, South Korea). [Oral communication]

2. **Han, H.**, Ulzijargal, B., Kim, S. (2023) The first record of *Thyas coronata* (Fabricius, 1775) (Lepidoptera: Noctuoidea: Erebidea) in Korean Peninsula. Spring Conference of Korean Society of Applied Entomology & The Entomological Society of Korea 2023 (Cheongju, South Korea). [Poster]

1. **Han, H.** & Kim, S. (2022) Discovery of Butterflies (Lepidoptera from Bach Ma National Park in Southern Vietnam. Autumn Conference of Korean Society of Applied Entomology 2022 (Gyeongju, South Korea). [Poster]

Fundings

As A Collaborator

13. Project for Predictive Surveillance of High-Risk Migratory Plant Pests, Animal and Plant Quarantine Agency 2026.05–2026.11

12. Transcriptome-Based Phylogenetic Analysis and Inference of Defense Mechanism Evolution in the Ditrysia (Lepidoptera), National Research Foundation of Korea 2026.04–2026.11

11. Region-Specific Monitoring of Forest Pests and Diseases (Jeolla Region), Korea Forestry Promotion Institute 2025.07–2025.11

10. Discovery of Indigenous Insect Species in Insular and Coastal Regions, Honam National Institute of Biological Resources 2025.04–2025.11

9. Transcriptome-Based Phylogenetic Analysis and Inference of Defense Mechanism Evolution in the Ditrysia (Lepidoptera), National Research Foundation of Korea 2025.04–2026.03

8. Monitoring Insect Pests and Plant Diseases on Fruit F lies Using an Automated Roll-Trap, Rural Development Administration 2025.04–2025.08

7. Transcriptome-Based Phylogenetic Analysis and Inference of Defense Mechanism Evolution in the Ditrysia (Lepidoptera), National Research Foundation of Korea 2024.04–2025.03

6. Honam National Institute of Biological Resources's Native Species Survey Project, Honam National Institute of Biological Resources 2024.04–2024.12

5. 2024 New and Emerging Pest Occurrence and Ecological Survey, Korea Forestry Promotion Institute 2024.04–2024.12

4. AI-based Plant Disease Monitoring and Precision Diagnosis Technology Development, Jeonbuk National University 2023.07–2023.12

3. AI-based Pest and Disease Management System, Korea Institute of Planning and Evaluation for Technology in Food, Agriculture and Forestry 2023.06–2023.12

2. Species delimitation, Gene-correlation and Biogeography of the genus *Promalactis* (Lepidoptera: Oecophoridae) based on Wing pattern evolution, National Research Foundation of Korea 2022.03–2023.05

1. Phylogeography of earthworm in enchytraeid group (Clitellate: Enchytraeidae), National Research Foundation of Korea 2021.04–2022.02

Skills

Technical Skills

Computational / Bioinformatics Skills

Phylogenetics & Evolution: IQ-TREE, MrBayes, BEAST

Sequence Analysis: SRA Toolkit, FastQC, Pipeline for Anchored Phylogenomics (Trimgalore!, USEARCH, Bridger, NCBI blastn, MEGAHIT, MAFFT, MACSE, FASconCAT-G)

Programming & Web: R, Python, GitHub Pages, Jekyll

Wet-Lab Skills

DNA Extraction, Gel Electrophoresis, PCR

Scientific Imaging & Design

Adobe Photoshop & Lightroom, Helicon focus, Inkscape, FigTree

Fieldwork Experience

Fieldwork in the Korean Peninsula (2018 – present): Conducted field surveys collecting Lepidoptera (focusing on Erebidae) using light traps, bucket traps, and sweeping nets to gather samples for phylogenomic analysis and taxonomic study.

Research Visit in Florida, USA (2025): Participated in a laboratory visit to co-work with senior researchers in the field of Lepidoptera phylogenomics and systematics.

Field Collection in Madagascar (2026): Participated in a collaborative collecting trip to gather samples for WGS and AHE sequencing.

Language

Korean: Native

English: CEFR C1 (TOEFL iBT 110 [March, 2026])

Research interests

1. Lepidoptera (butterflies and moths) and their evolutionary history
2. Integrative study using taxonomy, systematics, and genomics
3. Various perspectives from microevolution to macroevolution
4. Contribution to the society and Lepidoptera biodiversity and conservation